

ECOWAVE · FINAL DELIVERY · MAY 2026

UCD Process & Usability Validation

A mobile app to centralize citizen beach-cleaning brigades in Yucatán

Esteban Canto Vázquez · Eduardo José Matos Uc · Ángel Leandro Puch Uribe
Mauricio Emiliano Ramírez Ceciliano · Jorge Gabriel Vázquez Suárez
Software Engineering — Facultad de Matemáticas, UADY

What you'll see in 10 minutes

01

Problem & proposal

Why EcoWave and what it does

03

Requirements & critical case

8 RU, 10 RF, 11 NFR and CU-002

05

Heuristic evaluation

9 findings, 3 majors resolved

07

Usability test

6 users · SUS 82.8 · 100% success

02

UCD methodology

The seven-stage pipeline we ran

04

Prototypes

LF → MF → HF iterations

06

KLM analysis

12.0 s predicted for CU-002

08

Lessons & next iteration

What we keep and what we fix

Problem & proposal

PROBLEM

33 tons / year without a centralized channel

- Brigades are scattered across Facebook and WhatsApp groups.
- Volunteers have no visible incentives to stay engaged.
- Reports to environmental authorities are produced manually.

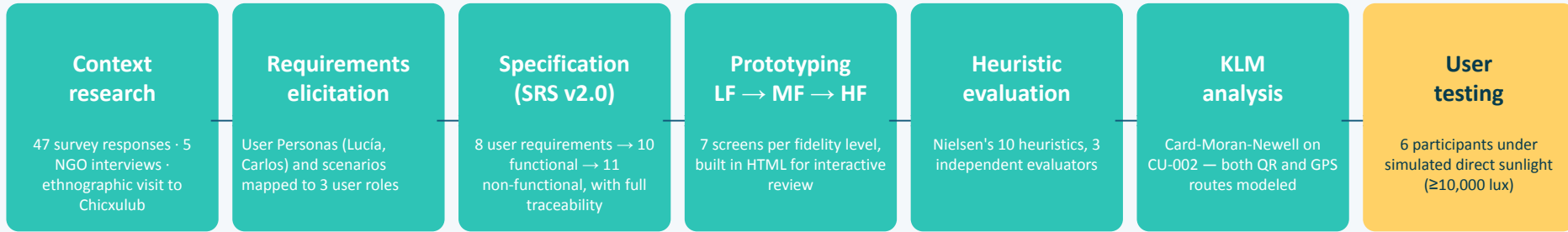
OUR PROPOSAL

A mobile platform with three pillars

- Interactive map of brigades within 25 km radius.
- Verified check-in: QR (primary) and GPS (fallback).
- Gamification with four badges (Bronze → Platinum).
- PDF impact reports for environmental authorities.

User-Centered Design pipeline

Seven sequential stages, each closed by a peer-reviewed artifact before moving on.



Traceability matrix kept every requirement aligned

8

User requirements (RU-01..RU-08)

10

Functional requirements (RF-01..RF-10)

11

Non-functional requirements (NFR-01..NFR-11)

CU-002 — Perform Check-in

The only requirement marked as Critical priority — concentrates four quantitative NFRs.

NFR-04

≤ 3 interactions

from the map screen to confirmation

NFR-05

≤ 30 s total

from app open to a confirmed check-in

NFR-06

≤ 500 ms feedback

visual response after every user action

NFR-07

WCAG AA accessibility

contrast $\geq 4.5:1$, taps $\geq 44 \times 44$ px, type ≥ 16 px

Three iterations · seven screens · one critical flow

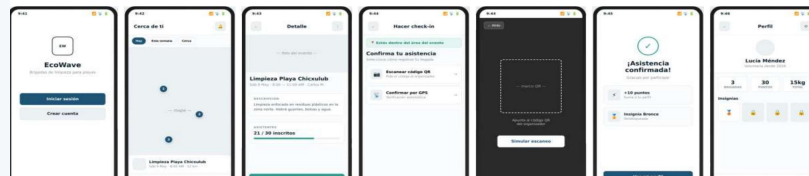
LF Architecture & flow

Black-and-white sketches to validate structure without color noise



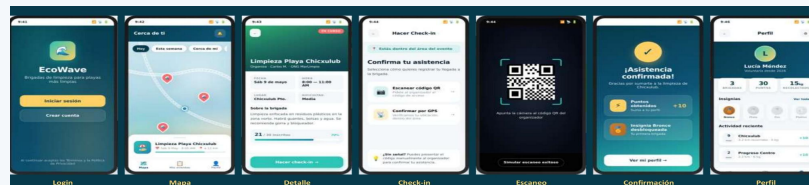
MF Hierarchy & components

Real proportions, neutral palette — focus on legibility



HF Final look & feel

Brand palette, microinteractions, complete states



9 findings · 3 evaluators · 5 already resolved

BY SEVERITY

3 / 3

Major findings resolved

1 / 4

Minor findings closed

1 / 2

Cosmetic findings closed

Three major findings — all closed before user testing

H-01

Visibility of QR scan state

Added animated scanning line + 'Point the camera at the code' message

H-06

Error prevention when outside the event area

Added 'You are inside the event area ✓' indicator before showing check-in options

H-09

Recovery from failed scan

Added 'No signal? Ask the organizer for the manual code' fallback block

KLM predicted 12.0 s for CU-002

Card, Moran & Newell — applied with Holleis et al. mobile extensions on the HF prototype.

PREDICTED TIME

12.0

seconds (QR route)

60% below the NFR-05 threshold of 30 s.

GPS route — even faster at 9.9 s.

Operator breakdown

Mental acts (M)	$5 \times 1.20 \text{ s}$	6.00 s
Taps (P)	$3 \times 1.10 \text{ s}$	3.30 s
Repositioning (H)	$1 \times 0.40 \text{ s}$	0.40 s
System response	variable	2.30 s

Total predicted

12.00 s

Test design — controlled and replicable



Objective

Validate that CU-002 (Perform Check-in) meets the four associated NFRs in a realistic simulated environment.



Participants

6 users recruited externally:

- 4 casual volunteers (19–34 y/o)
- 2 organizers (32 and 41 y/o)



Scenario

Arrive at a Chicxulub cleanup, register attendance, open the app, scan the QR, verify badge.



Instruments

Pre-test: profile · Post-test: SUS + SEQ
Observer: time, errors, taps, latency
LED lamp $\geq 10,000$ lux · screen recording

Results — SUS 82.8, all participants succeeded

18.3 s

Mean check-in time
Range 15.3–24.0 s · σ 3.5

100 %

Task success rate
5/6 unaided · 1/6 with prompt

82.8 /100

Mean SUS score
Excellent (Bangor band)

334 ms

Mean feedback latency
Range 291–429 ms

39% headroom between observed time and the NFR-05 threshold — enough margin to absorb real-world variability.

Per-participant breakdown

All six participants completed CU-002 — full quantitative data and outlier analysis.

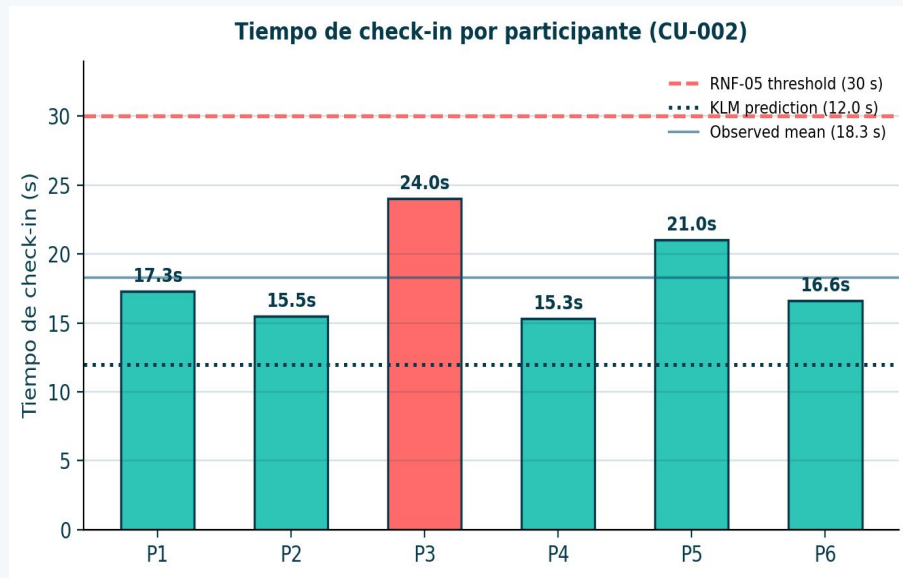


Fig. Check-in time per participant. Red line = NFR-05 threshold (30s); dotted black = KLM prediction; solid blue = observed mean.

INDIVIDUAL RESULTS

ID	Profile	Time	Taps	SUS	Aid
P1	Volunteer 22	17.3 s	3	82	—
P2	Volunteer 28	15.5 s	3	89	—
P3	Volunteer 34	24.0 s	5	67	✓
P4	Volunteer 19	15.3 s	3	87	—
P5	Organizer 41	21.0 s	4	81	—
P6	Organizer 32	16.6 s	3	91	—
Mean	—	18.3 s	3.5	82.8	

P3 outlier (24.0 s · SUS 67) is fully explained by glare under the simulated LED light — addressed by NFR-07 reinforcement in iteration 2.

Qualitative findings

Patterns surfaced by interview and observation that the heuristic evaluation had missed.

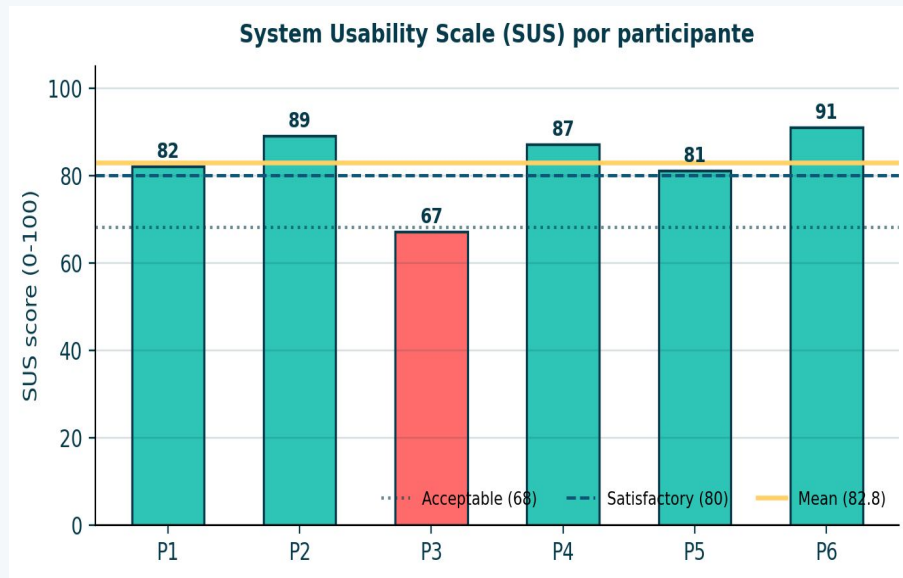


Fig. Individual SUS scores. Mean 82.8 lies in the Excellent band (Bangor et al.). P3 = LED-condition outlier.

FINDING 1 · LED REFLECTIONS

Light backgrounds suffer under simulated sunlight

4 of 6 participants tilted or shaded the screen during QR scan. One quoted: *“I could barely see the gray time text.”* Reinforces NFR-07 (contrast) in iteration 2.

FINDING 2 · LIGHT → DARK TRANSITION

Transient confusion at scanner screen

2 participants blinked and re-oriented when entering the dark QR scanner; one said: *“I thought it had turned off.”* Matches heuristic finding H-05 (cosmetic).

Bonus: 5 of 6 participants spontaneously asked how to unlock the next badge — gamification works.

What we learned from real users

WHAT WORKED

KLM predicted user behavior accurately

- Observed time (18.3 s) is 52% above KLM prediction — consistent with the 30–50% range reported by literature for novice users.
- Resolving the 3 major heuristic findings before user testing freed the session to surface finer issues.
- Gamification motivated 5 of 6 participants — two spontaneously asked how to unlock the next badge.

WHAT'S NEXT — ITERATION 2

Sunlight legibility is the priority

- Boost contrast of secondary information on light-background screens (NFR-07 stricter).
- Soften the transition between light and dark screens (cosmetic finding H-05).
- Close the four remaining heuristic findings — all of minor or cosmetic severity.
- Move from prototype to MVP development.

Five people · one process · equal commitment

Each member led one DCU stage and contributed across all others.



Esteban Canto Vázquez

Lead — Research & requirements



Eduardo José Matos Uc

Lead — High-fidelity prototype



Ángel Leandro Puch Uribe

Lead — Heuristic evaluation



Mauricio Emiliano Ramírez Ceciliano

Lead — KLM analysis



Jorge Gabriel Vázquez Suárez

Lead — Usability test protocol



Thank you.

Questions are welcome — and welcomed.

12.0 s

KLM

82.8

SUS

100%

Success